1 GROUND FLOOR PLAN

Notes

Author

Arnold Okema

Project

Office HVAC

Date

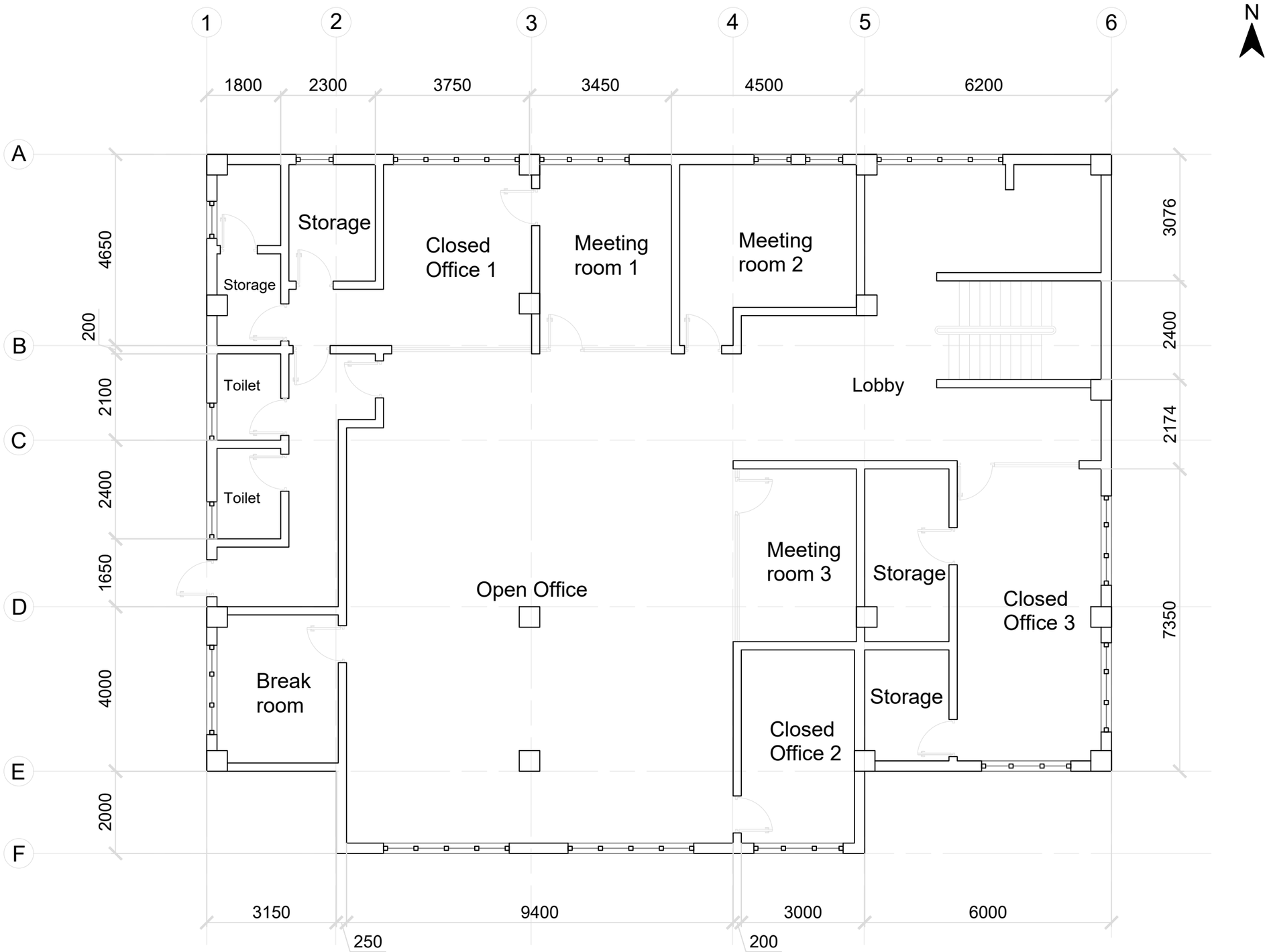
08/03/2026

Scale

1:100

Sheet

A101



1 FIRST AND SECOND FLOOR PLAN

Notes

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Date

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1:100

Sheet

A102

1 SOUTH ELEVATION

Notes

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Scale

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Sheet

A201

1 WEST ELEVATION

Notes

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Scale

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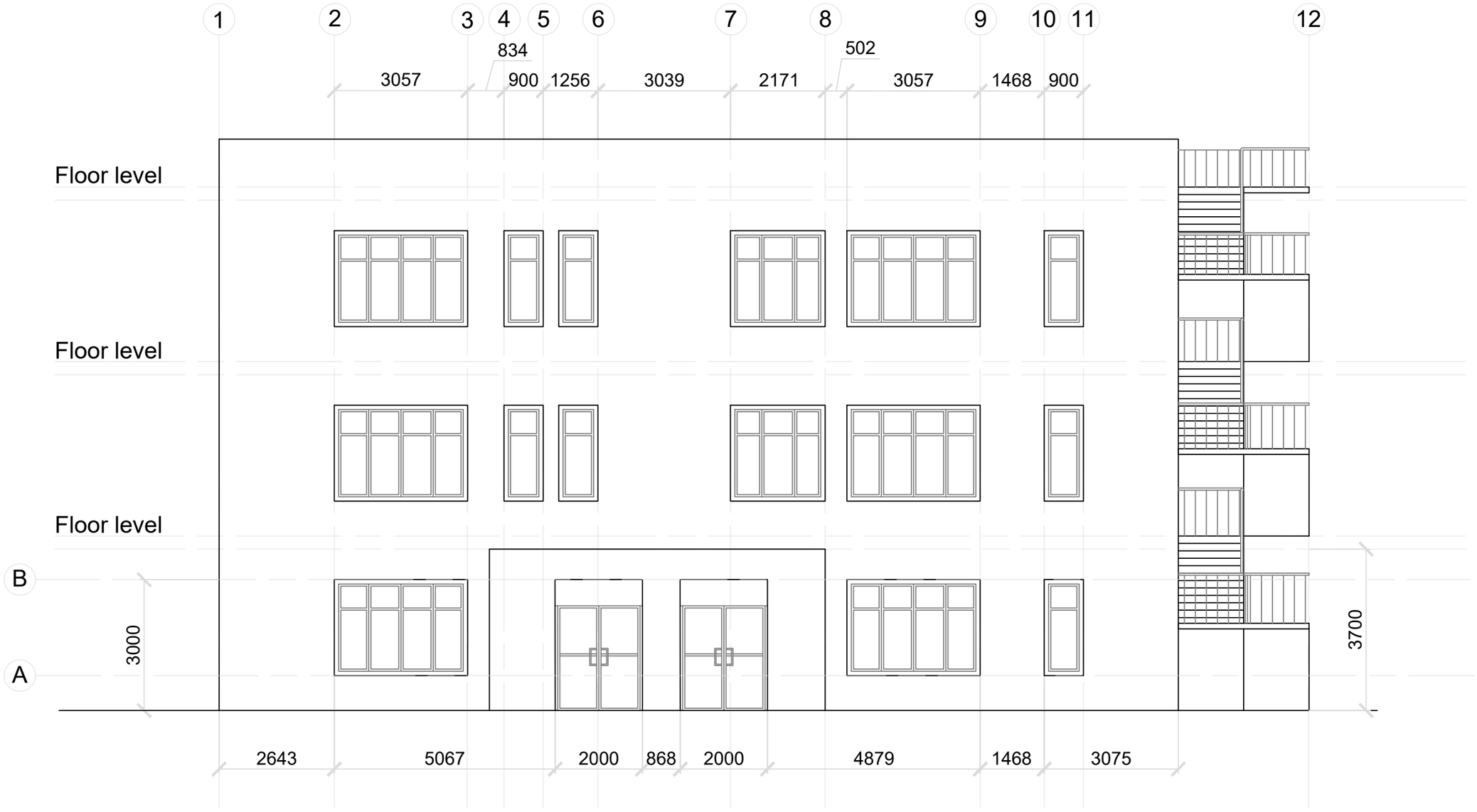
Sheet

A202



1 NORTH ELEVATION

Notes

Author

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Date

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1:100

Sheet

A203

1 EAST ELEVATION

Notes

Author

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Date
08/03/2026

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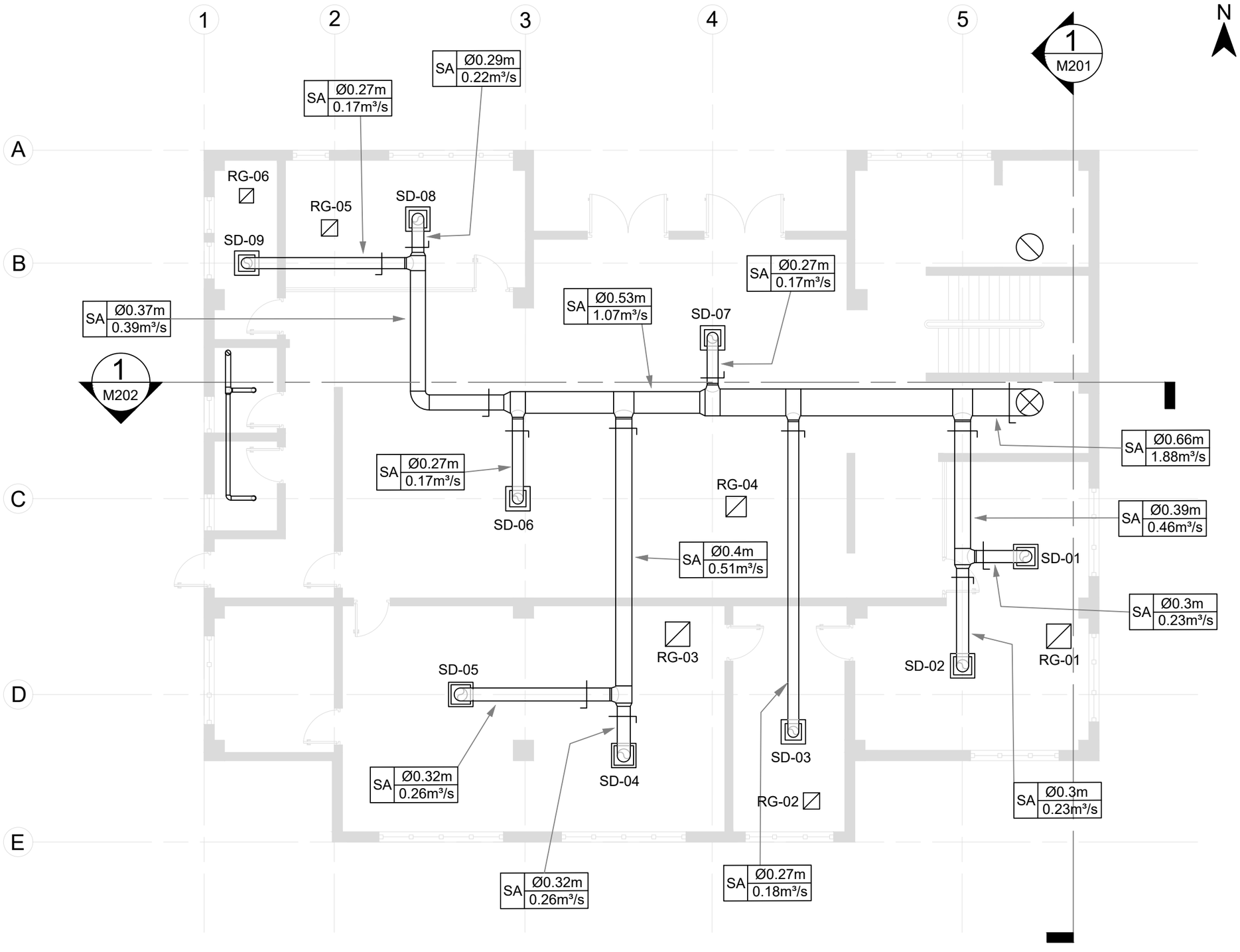
1:100

Sheet
A204



1 GROUND FLOOR HVAC LAYOUT

Notes

Author

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Date
08/03/2026

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1:100

Sheet
M101

GENERAL NOTES:

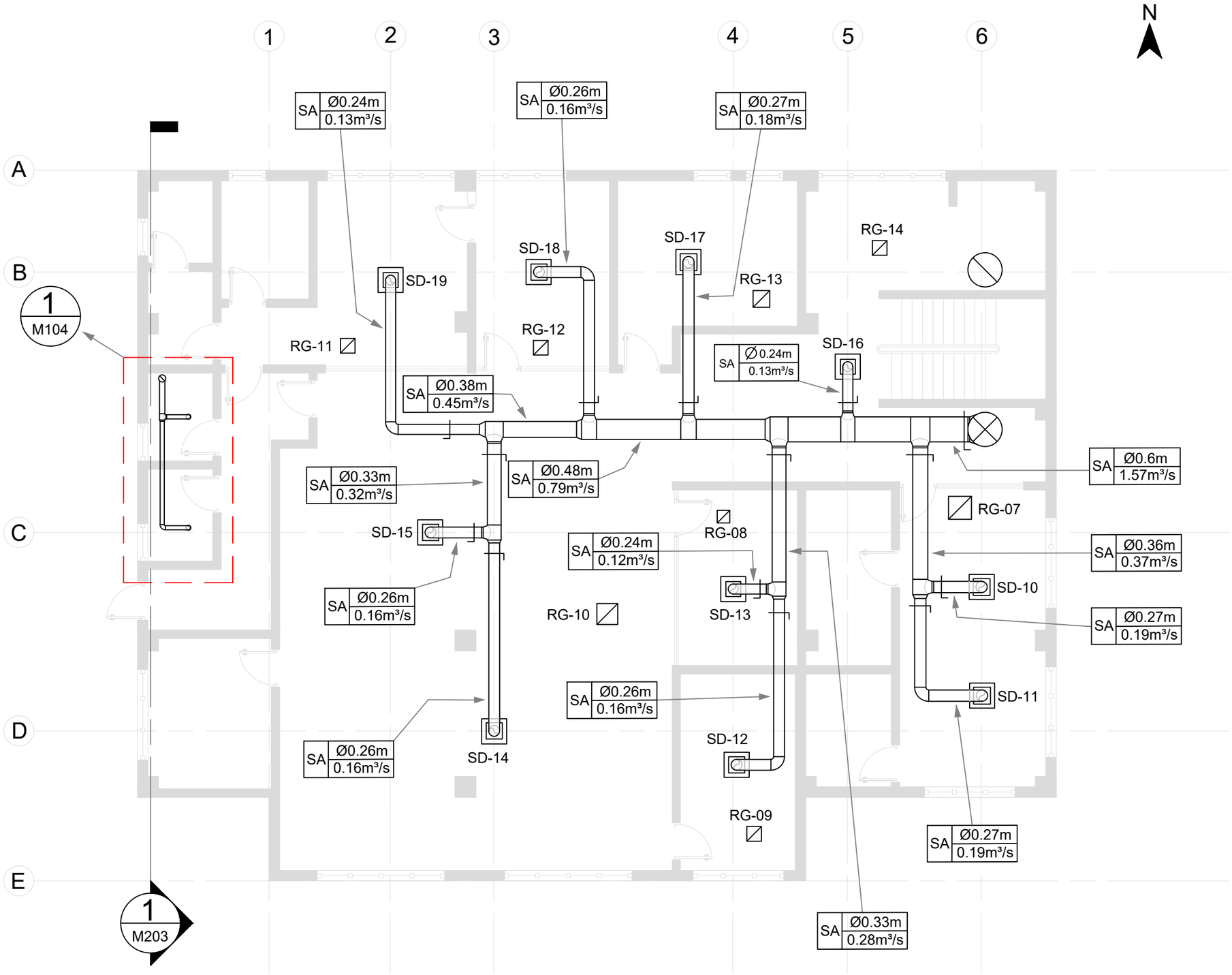
1. Intermittently occupied spaces including toilets, corridors, and break rooms are not provided with dedicated supply air terminals.

2. These spaces are ventilated through transfer air from adjacent conditioned spaces via door undercuts and internal partition openings.

3. To facilitate return air circulation, openings are provided at ceiling plenum level above internal doors and glazed partitions.

4. The ceiling void functions as a return air plenum, allowing return air to flow freely toward the return riser serving the rooftop unit.

5. The arrangement ensures adequate air movement between spaces while maintaining the required negative pressure in toilet areas relative to adjacent occupied zones.



1 FIRST AND SECOND FLOOR HVAC LAYOUT

Notes

Author

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1:100

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M102

GENERAL NOTES:

1. The calculated index run pressure loss is 97.6Pa.

2. 10% was added as a safety margin to account for installation variability.

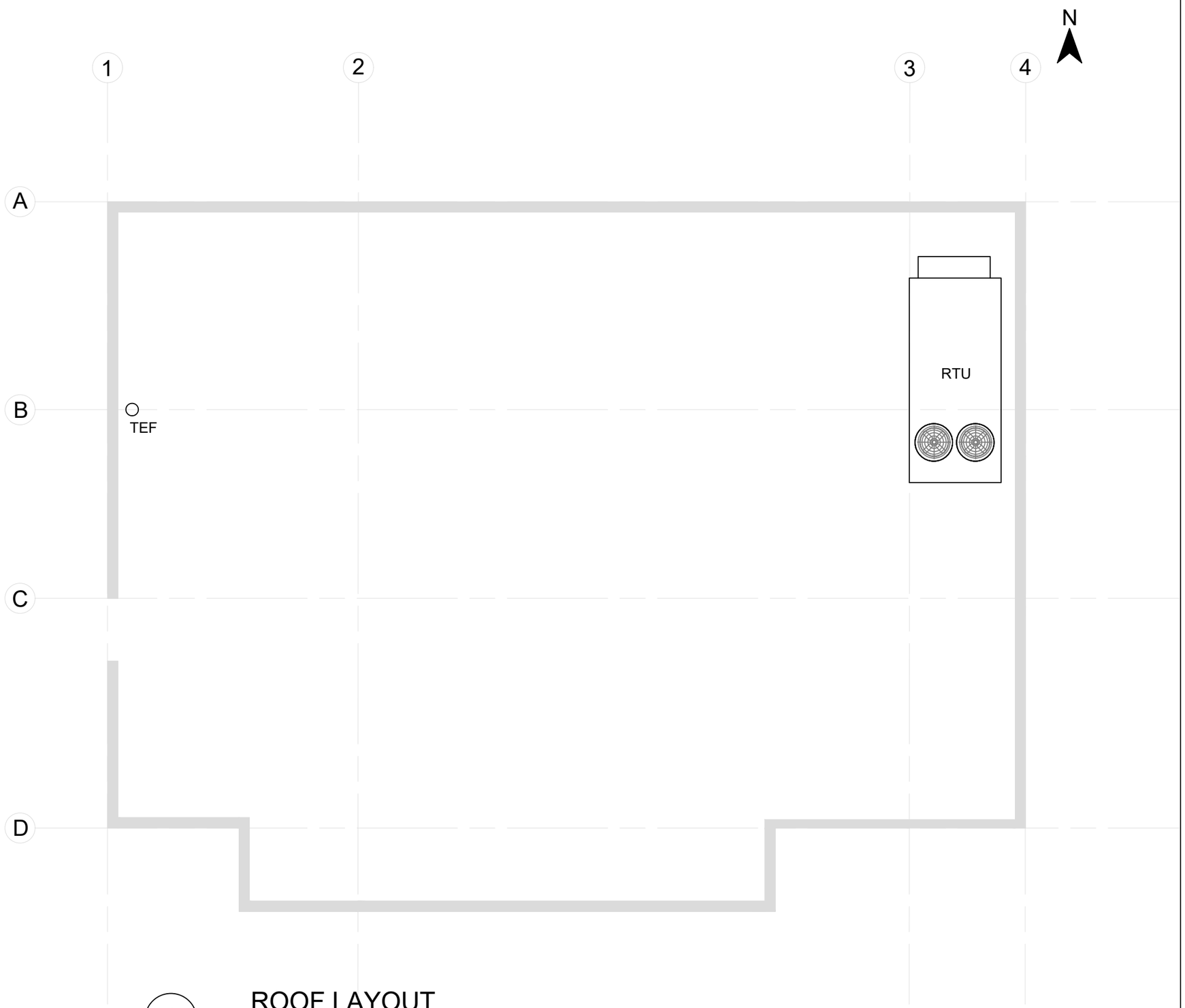
3. Final calculated index run pressure loss is 107.4Pa. Value rounded up to 110Pa.

4. RTU fan External Static Pressure is 110Pa.

5. RTU components assumed to cause internal static pressure of 200Pa.

6. Total RTU fan static pressure is 310Pa.

7. 2.4KW fan power calculated, based on 70% assumed efficiency.



1 ROOF LAYOUT

Notes

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1:100

Sheet
M103

GENERAL NOTES:

1. Each toilet is provided with mechanical extract ventilation at a rate of 0.015 m³/s per toilet.

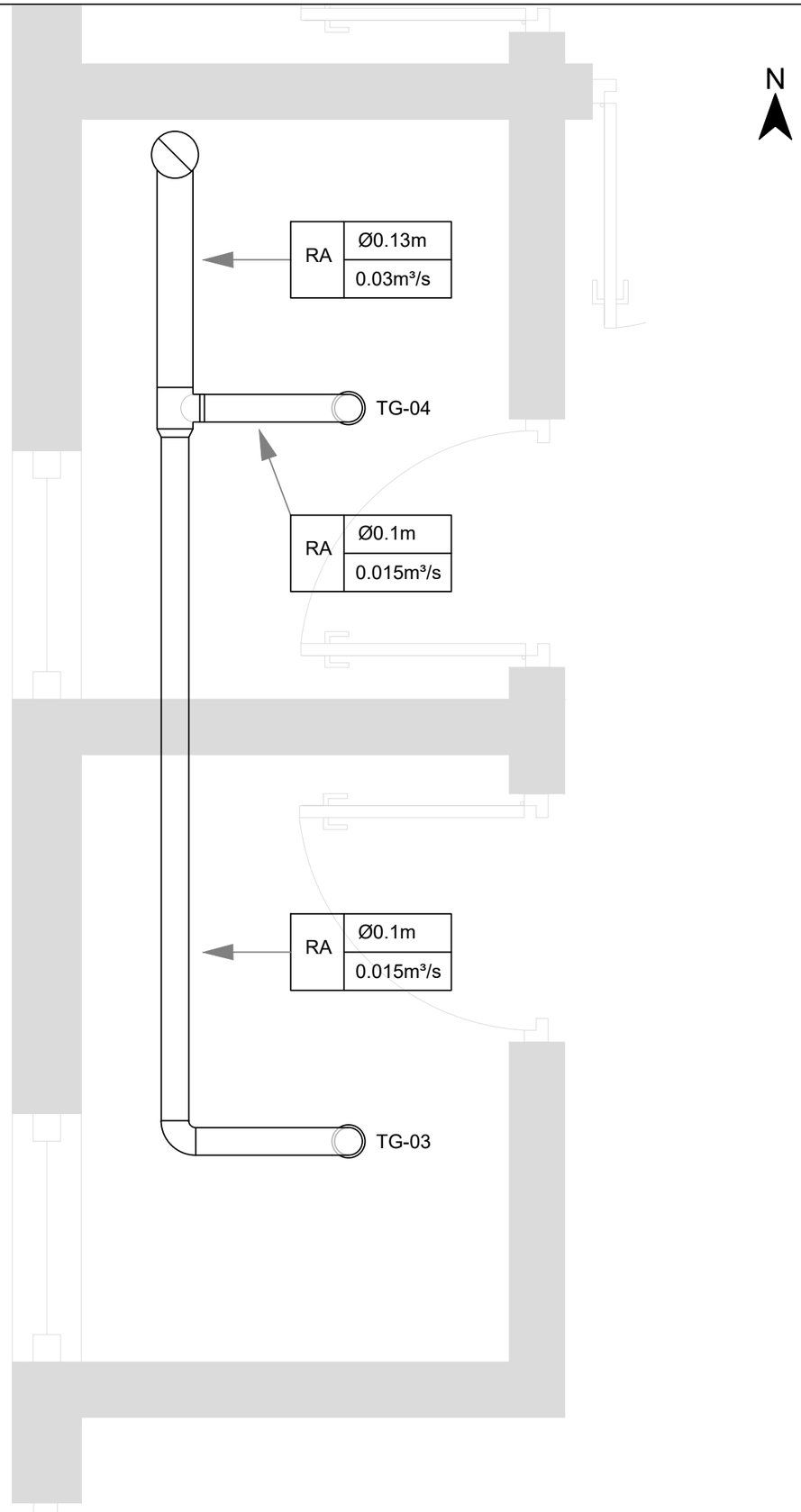
2. The extract system is designed to maintain toilet spaces under negative pressure relative to adjacent occupied areas, preventing odour migration and cross-contamination.

3. Extract air is conveyed through dedicated ductwork to a roof-mounted extract fan.

4. The fan discharges air to atmosphere at roof level.

5. No return or recirculated air from toilet spaces is permitted. All extracted air is discharged directly to atmosphere.

6. Replacement air is provided via transfer air from adjacent spaces through door undercuts.



1 ENLARGED TOILET HVAC LAYOUT

Notes

Author

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Date
08/03/2026

Scale
1:20

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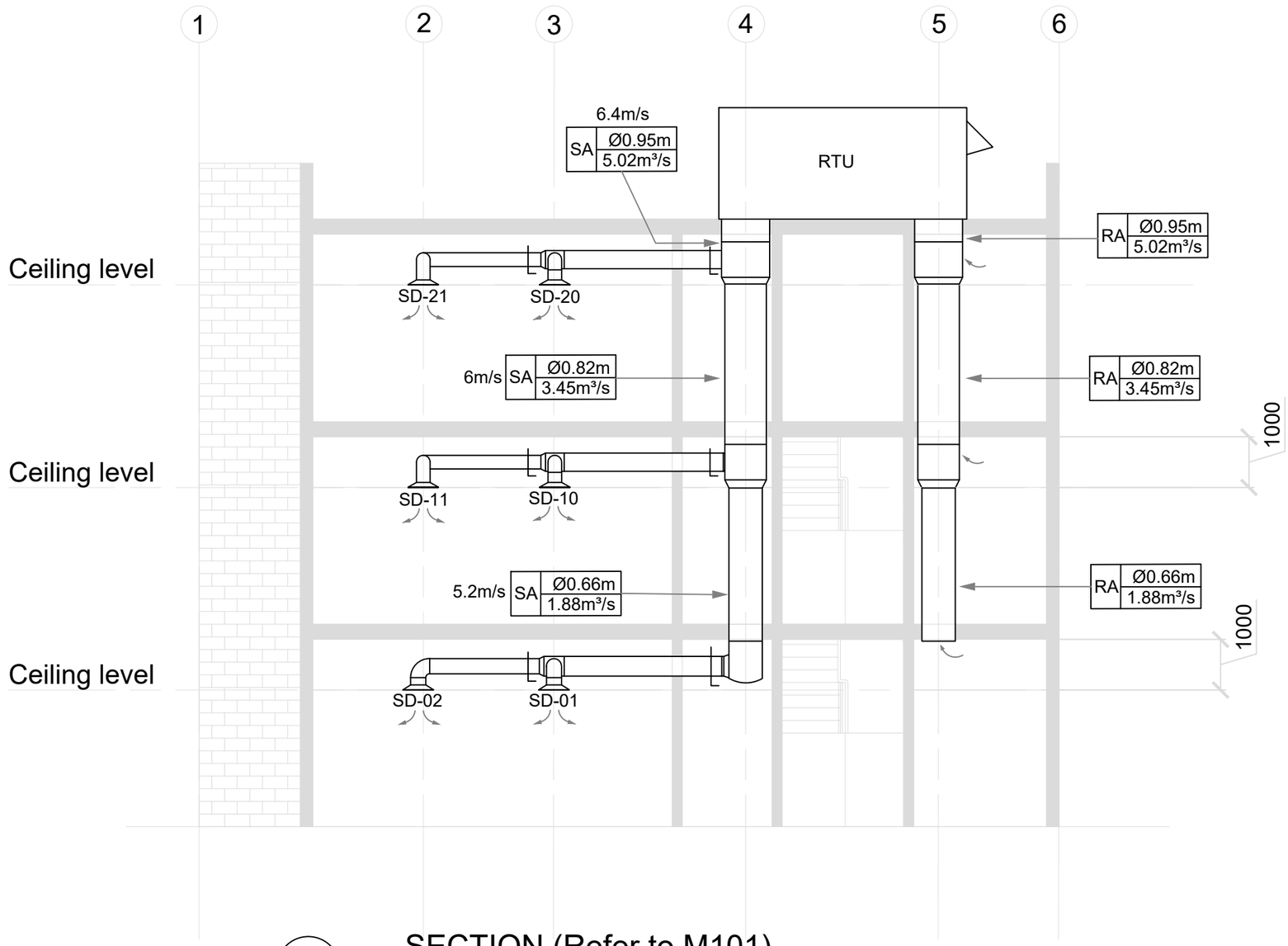
M104

GENERAL NOTES:

1. The ceiling void is used as a plenum to return air to the RTU.

2. The negative pressure created by the RTU fan sucks the return air from the conditioned spaces up into the plenum.

3. The air then flows to the return riser and finds its way back to the RTU where it is either recirculated or expelled into the atmosphere.



1 SECTION (Refer to M101)

Notes

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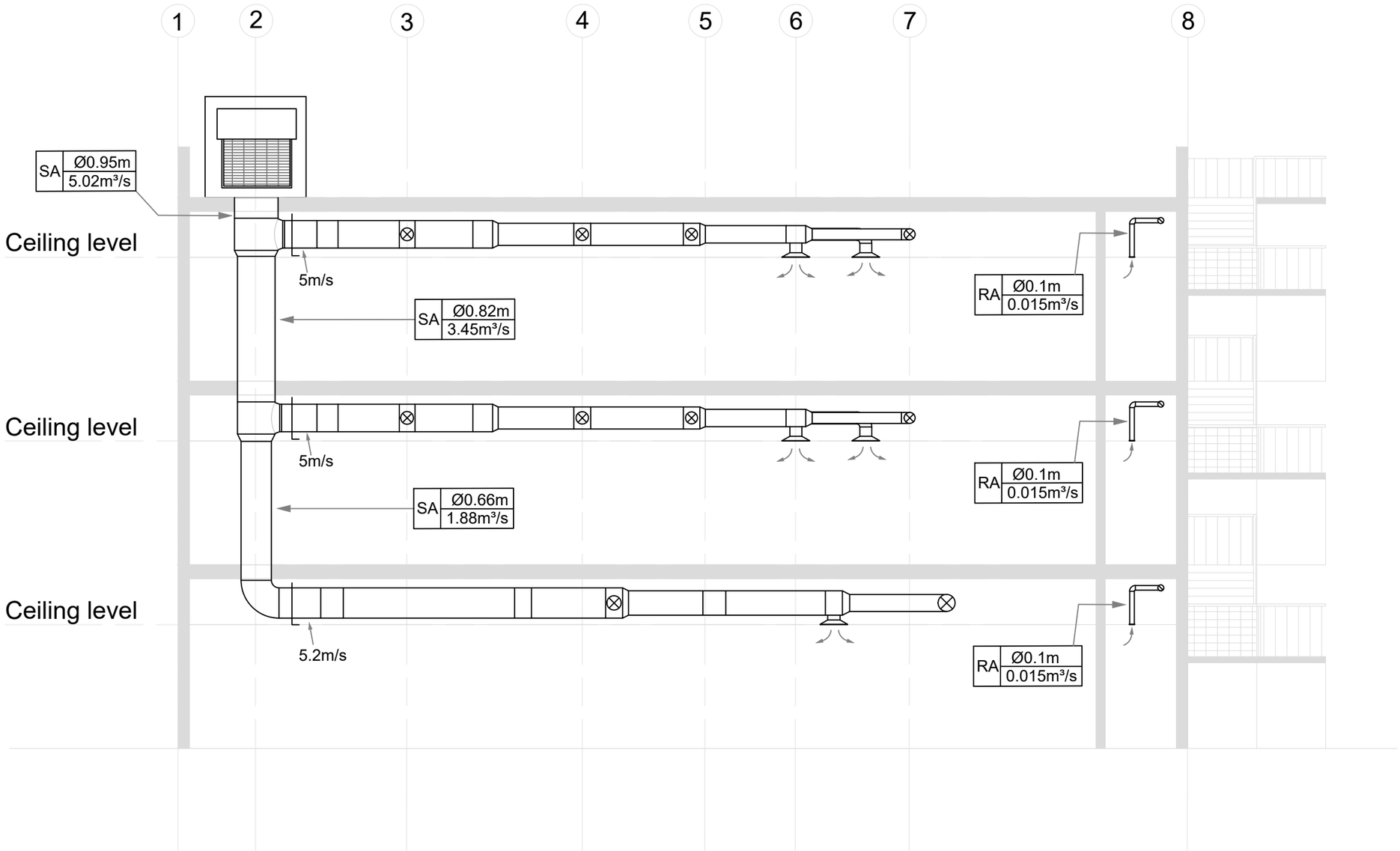
M201



1 SECTION (Refer to M101)

Notes

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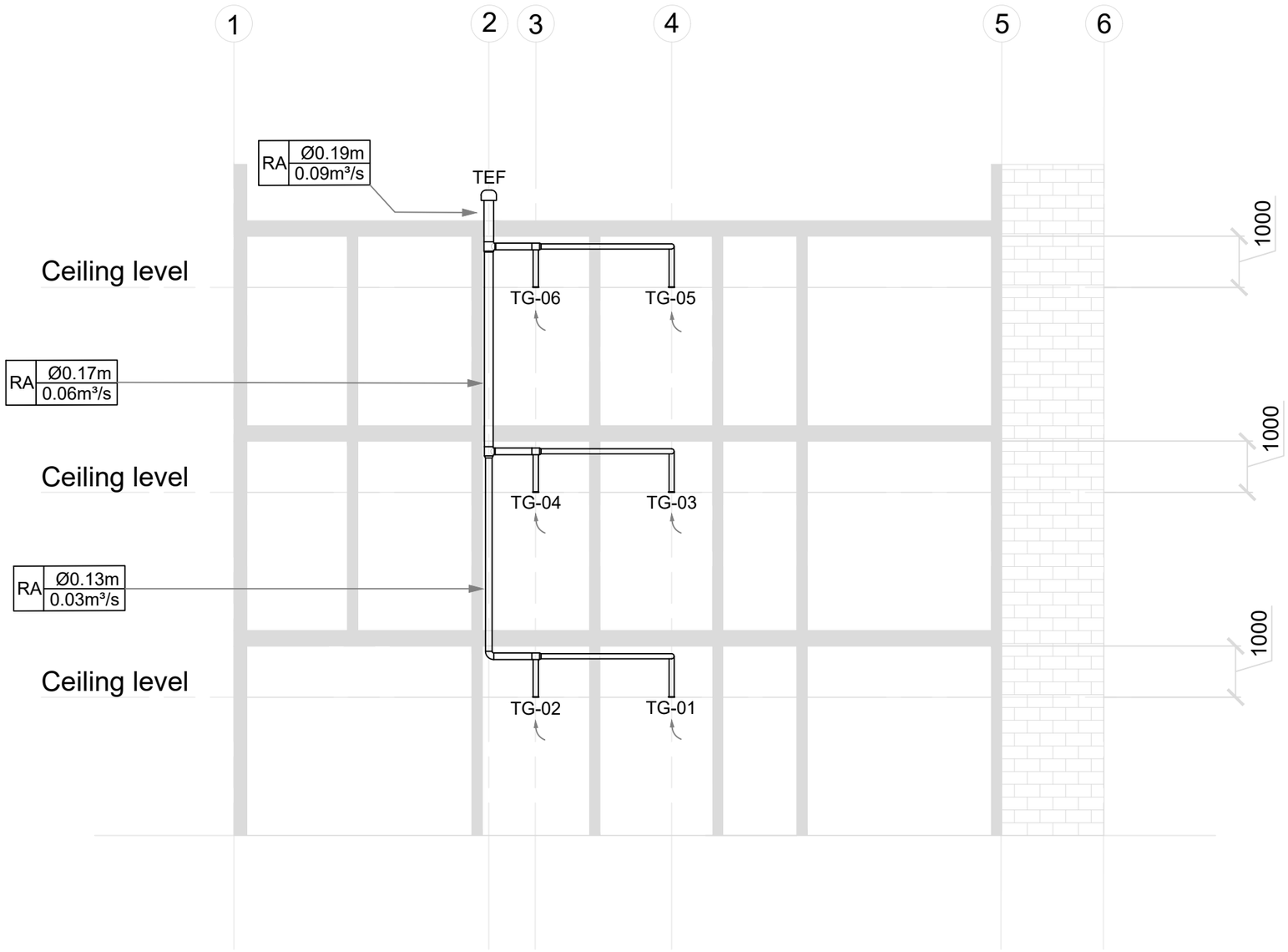
1:100

Sheet
M202



1 SECTION (Refer to M102)

Notes

Author

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1:100

Sheet
M203

SUPPLY DIFFUSER SCHEDULE

TAG	TYPE	DIMENSIONS (mm)	NECK SIZE (mm)	AIR FLOW (m^3/s)	AIR VELOCITY (m/s)	FLOOR	LOCATION
SD-01	4-Way ceiling diffuser	600x600	300	0.23	3.2	Ground	Office2
SD-02	4-Way ceiling diffuser	600x600	300	0.23	3.2	Ground	Office2
SD-03	4-Way ceiling diffuser	600x600	270	0.18	2.9	Ground	MeetingRM2
SD-04	4-Way ceiling diffuser	600x600	320	0.26	3.2	Ground	Office1
SD-05	4-Way ceiling diffuser	600x600	320	0.26	3.2	Ground	Office1
SD-06	4-Way ceiling diffuser	600x600	270	0.17	2.8	Ground	Lobby
SD-07	4-Way ceiling diffuser	600x600	270	0.17	2.8	Ground	Lobby
SD-08	4-Way ceiling diffuser	600x600	290	0.22	3.2	Ground	MeetingRM1
SD-09	4-Way ceiling diffuser	600x600	270	0.17	2.8	Ground	Lounge
SD-10	4-Way ceiling diffuser	600x600	270	0.19	2.9	First	Office 3
SD-11	4-Way ceiling diffuser	600x600	270	0.19	2.9	First	Office 3
SD-12	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	Office 2
SD-13	4-Way ceiling diffuser	600x600	240	0.12	2.7	First	MeetingRM3
SD-14	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	OpenOffice
SD-15	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	OpenOffice
SD-16	4-Way ceiling diffuser	600x600	240	0.13	2.7	First	Lobby
SD-17	4-Way ceiling diffuser	600x600	270	0.18	2.9	First	MeetingRM2
SD-18	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	MeetingRM1
SD-19	4-Way ceiling diffuser	600x600	240	0.13	2.7	First	Office1
SD-20	4-Way ceiling diffuser	600x600	270	0.19	2.9	Second	Office 3
SD-21	4-Way ceiling diffuser	600x600	270	0.19	2.9	Second	Office 3
SD-22	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	Office 2
SD-23	4-Way ceiling diffuser	600x600	240	0.12	2.7	Second	MeetingRM3
SD-24	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	OpenOffice
SD-25	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	OpenOffice
SD-26	4-Way ceiling diffuser	600x600	240	0.13	2.7	Second	Lobby
SD-27	4-Way ceiling diffuser	600x600	270	0.18	2.9	Second	MeetingRM2
SD-28	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	MeetingRM1
SD-29	4-Way ceiling diffuser	600x600	240	0.13	2.7	Second	Office1

MECHANICAL LEGEND	
SA	Supply air
RA	Return air
SD	Supply diffuser
RG	Return grille
TG	Toilet grille
RTU	Rooftop unit
TEF	Toilet extract fan
	Volume control damper
	Supply duct section
	Return duct section
	Section marker
	Enlarged detail marker

Notes

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08/03/2026

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1:100

Sheet

M301

RETURN GRILLE SCHEDULE

TAG	TYPE	DIMENSIONS (mm)	AIR FLOW (m³/s)	MOUNTING	FLOOR	LOCATION
RG-01	Eggcrate grille	600x600	0.46	Ceiling	Ground	Office 2
RG-02	Eggcrate grille	400x400	0.18	Ceiling	Ground	MeetingRM2
RG-03	Eggcrate grille	600x600	0.51	Ceiling	Ground	Office 1
RG-04	Eggcrate grille	500x500	0.34	Ceiling	Ground	Lobby
RG-05	Eggcrate grille	400x400	0.22	Ceiling	Ground	MeetingRM1
RG-06	Eggcrate grille	350x350	0.17	Ceiling	Ground	Lounge
RG-07	Eggcrate grille	550x550	0.37	Ceiling	First	Office 3
RG-08	Eggcrate grille	300x300	0.12	Ceiling	First	MeetingRM3
RG-09	Eggcrate grille	350x350	0.16	Ceiling	First	Office 2
RG-10	Eggcrate grille	500x500	0.32	Ceiling	First	OpenOffice
RG-11	Eggcrate grille	350x350	0.13	Ceiling	First	Office 1
RG-12	Eggcrate grille	350x350	0.16	Ceiling	First	MeetingRM1
RG-13	Eggcrate grille	400x400	0.18	Ceiling	First	MeetingRM2
RG-14	Eggcrate grille	350x350	0.13	Ceiling	First	Lobby
RG-15	Eggcrate grille	550x550	0.37	Ceiling	Second	Office 3
RG-16	Eggcrate grille	300x300	0.12	Ceiling	Second	MeetingRM3
RG-17	Eggcrate grille	350x350	0.16	Ceiling	Second	Office 2
RG-18	Eggcrate grille	500x500	0.32	Ceiling	Second	OpenOffice
RG-19	Eggcrate grille	350x350	0.13	Ceiling	Second	Office 1
RG-20	Eggcrate grille	350x350	0.16	Ceiling	Second	MeetingRM1
RG-21	Eggcrate grille	400x400	0.18	Ceiling	Second	MeetingRM2
RG-22	Eggcrate grille	350x350	0.13	Ceiling	Second	Lobby

TOILET GRILLE SCHEDULE

TAG	TYPE	DIMENSIONS (mm)	AIR FLOW (m³/s)	MOUNTING	FLOOR
TG-01	Circular	Ø100	0.015	Ceiling	Ground
TG-02	Circular	Ø100	0.015	Ceiling	Ground
TG-03	Circular	Ø100	0.015	Ceiling	First
TG-04	Circular	Ø100	0.015	Ceiling	First
TG-05	Circular	Ø100	0.015	Ceiling	Second
TG-06	Circular	Ø100	0.015	Ceiling	Second

RTU SCHEDULE

TAG	UNIT TYPE	SUPPLY AIR FLOW (m³/s)	EXTERNAL STATIC PRESSURE (Pa)	COOLING CAPACITY (KW)	HEATING CAPACITY (KW)	FAN POWER (KW)	LOCATION
RTU	Packaged Rooftop	5.02	110	62	77.3	2.4	Roof

TOILET FAN SCHEDULE

TAG	FAN TYPE	AIR FLOW (m³/s)	EXTERNAL STATIC PRESSURE (Pa)	FAN POWER (KW)	LOCATION
TEF	Centrifugal Inline	0.09	40	0.0065	Roof

Notes

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08/03/2026

Scale
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Sheet
M302